Multiple matrix transformations

Key Points

When performing matrix transform A followed by B we do BA

That is successive matrix transforms are applied as a premultiplier

Basic Skills

Q1. Compute the following matrix calculations

a) $\begin{pmatrix} 3 & -1 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} -2 & 2 \\ 1 & 0 \end{pmatrix}$ **b)** $\begin{pmatrix} -1 & 4 \\ 3 & 0 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 3 & 6 \end{pmatrix}$

c) $\begin{pmatrix} 2 & 2 & 6 \\ 1 & -1 & 4 \\ 0 & 2 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 & -5 \\ 0 & 2 & 3 \\ -4 & 1 & 4 \end{pmatrix}$

Q2. Write a matrix for the following 2D transforms

a) A stretch in y by a scale factor of 4

b) A clockwise rotation of 30°

Q3. Write a matrix for the following 3D transforms

a) A reflection in $z = 0$

b) Anticlockwise around x-axis by 30°

Competency Skills

Q4. Find the matrix which first stretch's the x-coordinates by scale factor 4 and then stretch's the y-coordinates by scale factor 3

Q5. Write the matrix transformation for a reflection in $y = -x$ followed by the shear given by $\begin{pmatrix} 5 & 0 \\ 2 & 5 \end{pmatrix}$

Q6. In 3D find the matrix transform which first stretches the x-coordinates by scale factor 4 and then performs a reflection in the plane $z = 0$

Q7. Find the matrix for a reflection in $y = \frac{1}{\sqrt{3}}x$ followed by $y = \sqrt{3}x$

Q8. The matrix $\begin{pmatrix} 0 & 3 \\ -1 & 0 \end{pmatrix}$ represents two single transform S followed by transform T, find S and T

Q9. Write down a matrix for a reflection in $x = 0$ followed by $y = 0$ followed by $z = 0$

Advanced Skills

Q10. Prove that an anticlockwise rotation of θ° followed by a clockwise rotation of ϕ° is equivalent to an anticlockwise rotation of $(\theta - \phi)^\circ$

Q11. For the transform M which reflections coordinates in the line $y = mx$

a) Write $\cos 2\theta$ and $\sin 2\theta$ in terms of $\tan\theta$

b) Hence show that $M = \frac{1}{1+m^2}\begin{pmatrix} 1 - m^2 & 2m \\ 2m & m^2 - 1 \end{pmatrix}$

c) Prove that $M^n = \begin{cases} I_2 \ if \ n \ is \ an \ odd \ integer \\ M \ if \ n \ is \ an \ even \ integer \end{cases}$

Extension

Q12. Explain why we must premultiply by successive transforms and not postmultiply